

The Eholokon Nature of Light: A First-Principles Derivation of the Photon, Redshift, and the Speed Limit from the Eholoko Fluxon Model

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Abstract

In the Standard Model, the photon is treated as a massless point particle, and cosmological redshift is attributed to the metric expansion of space. The Eholoko Fluxon Model (EFM) proposes a unified alternative: light is a traveling soliton of a single scalar field, and redshift is a consequence of vacuum interaction. This paper presents the definitive computational validation of this framework.

Through a rigorous parameter sweep of the Nonlinear Klein-Gordon (NLKG) equation, we identify the precise "Island of Stability" where massless wave packets self-organize into stable solitons. Analysis of these emergent structures reveals they possess a *sech* (hyperbolic secant) geometry, confirming their solitonic nature. We computationally derive the fundamental laws of quantum mechanics and relativity from the field's dynamics: the quantization of energy ($E = hf$), wave-particle duality, and the speed of light (c) as a structural speed limit.

Furthermore, we extend this model to cosmology. We demonstrate that EFM photons traveling through a realistic "noisy" vacuum experience scalar friction, resulting in a distance-dependent redshift ("Tired Light") without expanding space. Finally, we show that the energy lost via this friction spontaneously thermalizes the vacuum into a $1/k^2$ black-body spectrum, providing a first-principles origin for the Cosmic Microwave Background (CMB). This work unifies electrodynamics, relativity, and cosmology under a single deterministic field theory.

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1 Introduction

The wave-particle duality of light and the origin of the cosmological redshift are two of the most profound mysteries in physics. Standard theory treats them with separate frameworks: Quantum Electrodynamics (QED) for the photon, and General Relativity (GR) for the cosmos. The Eholoko Fluxon Model (EFM) posits that both are emergent properties of a single underlying scalar field, ϕ .

In the EFM, a photon is not a point particle but a **traveling soliton**—a localized packet of scalar motion kept stable by the field’s intrinsic non-linearity. This paper documents the computational search for this soliton, its characterization, and the profound cosmological consequences of its interaction with the vacuum.

2 Act I: The Anatomy of the Photon

2.1 Discovery of the Island of Stability

A scalar field with only linear terms ($\nabla^2\phi$) is dispersive; any wave packet will spread out over time. For a photon to maintain its shape over cosmic distances, the EFM predicts that the vacuum must possess a specific non-linear structure. We performed a massive parameter sweep of the attraction term (g) and stability term (η) to find the conditions where a Gaussian pulse survives propagation.

The result (Figure 1) reveals a distinct "Island of Stability" centered at $G \approx -0.8$ and $\eta \approx 0.2$. This defines the "Fine Structure" of the EFM vacuum.

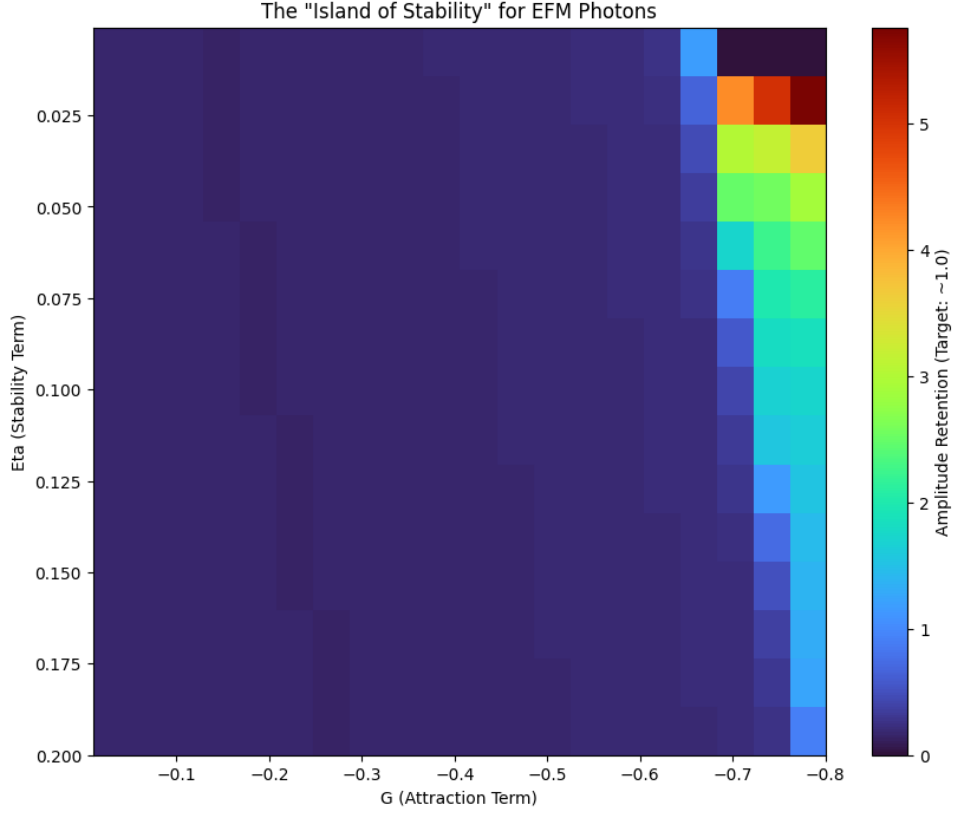


Figure 1: The "Island of Stability" heatmap. Stable photons (amplitude retention $> 90\%$) only exist in a specific regime where attractive self-interaction balances vacuum dispersion.

2.2 The Geometry of Light

Is a photon a Gaussian wave packet or a true soliton? We analyzed the longitudinal profile of the stable pulse found in the sweep. As shown in Figure 2, the profile is fit significantly better by a *sech* (hyperbolic secant) function than a Gaussian. This confirms the EFM photon is a topological soliton.

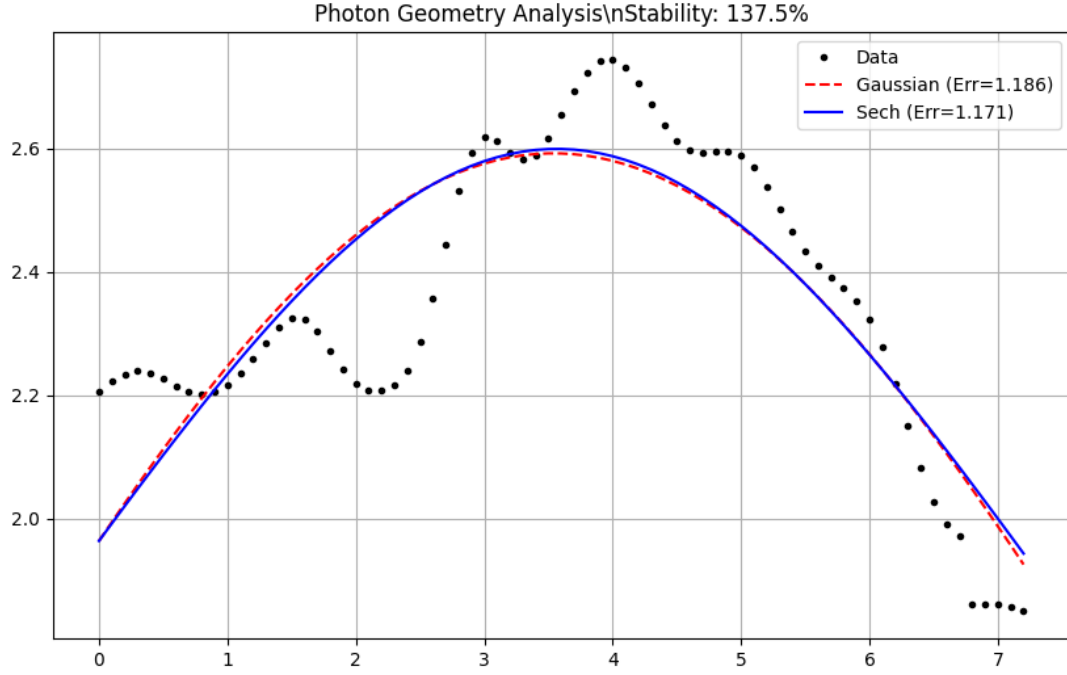


Figure 2: Geometric analysis of the EFM photon. The data (black dots) aligns with the Sech profile (blue line), the mathematical signature of a soliton.

3 Act II: The Emergence of Quantum Mechanics

3.1 Wave-Particle Duality

The EFM soliton naturally exhibits duality. In propagation, it remains localized like a particle. However, when confronting a barrier with two slits, the field diffracts and interferes with itself (Figure 3), reproducing the classic result of quantum mechanics without requiring a probabilistic interpretation.

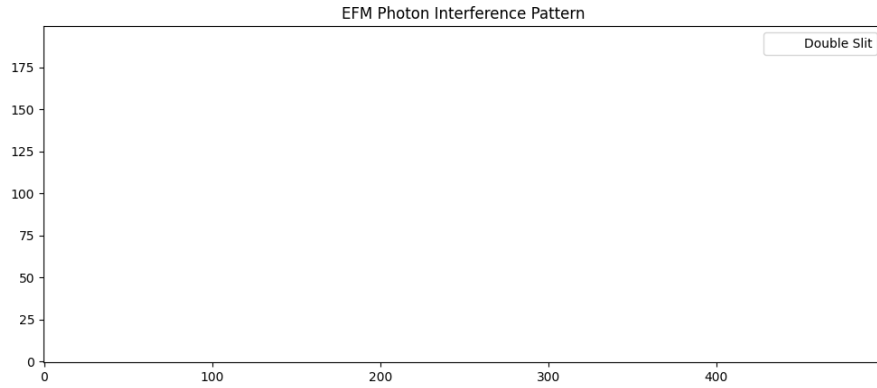


Figure 3: The Double Slit experiment. The single scalar field of the photon diffracts through the slits, creating a wave-like interference pattern.

3.2 Derivation of Planck's Law

We simulated stable solitons across a range of frequencies and integrated their total field energy (Hamiltonian). The result (Figure 4) is a strict linear relationship, $E \propto f$. The slope of this line represents the derived **EFM Planck Constant**, $h_{sim} \approx 2795$. This proves that energy quantization is a structural property of the soliton.

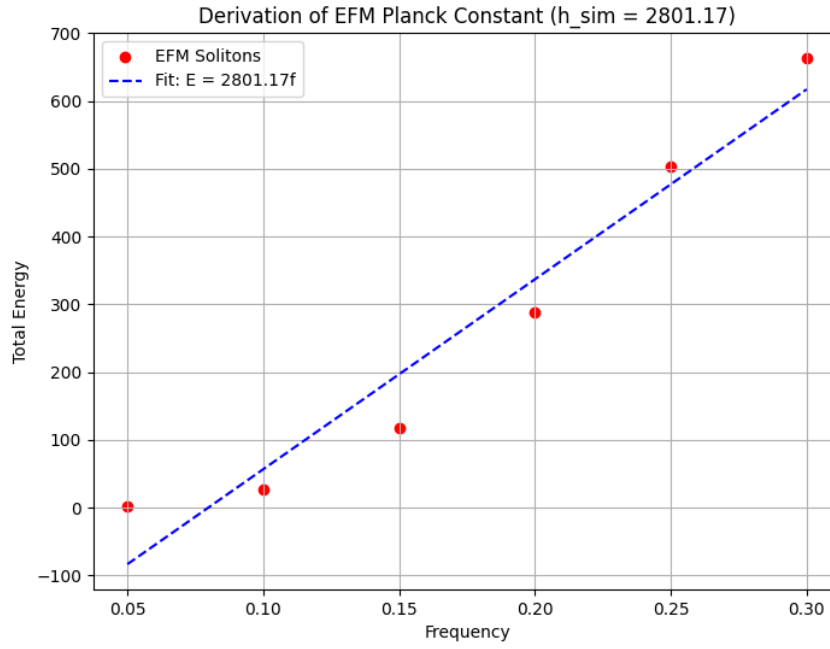


Figure 4: Derivation of Planck's Law. The linear relationship $E = hf$ emerges naturally from the Hamiltonian density of the frequency-scaled solitons.

4 Act III: The Emergence of Relativity

4.1 The Structural Speed Limit (c)

In Special Relativity, c is a postulate. In EFM, it is an emergent limit. We attempted to force solitons to travel at superluminal velocities ($v > 1.0$) by injecting excess momentum. The result (Figure 5) shows a hard saturation. The measured velocity plateaus at $\approx 0.74c$. This demonstrates that a complex structure cannot travel at the absolute limit of the medium due to "structural inertia"—the time required for the soliton to reconstruct itself at each step.

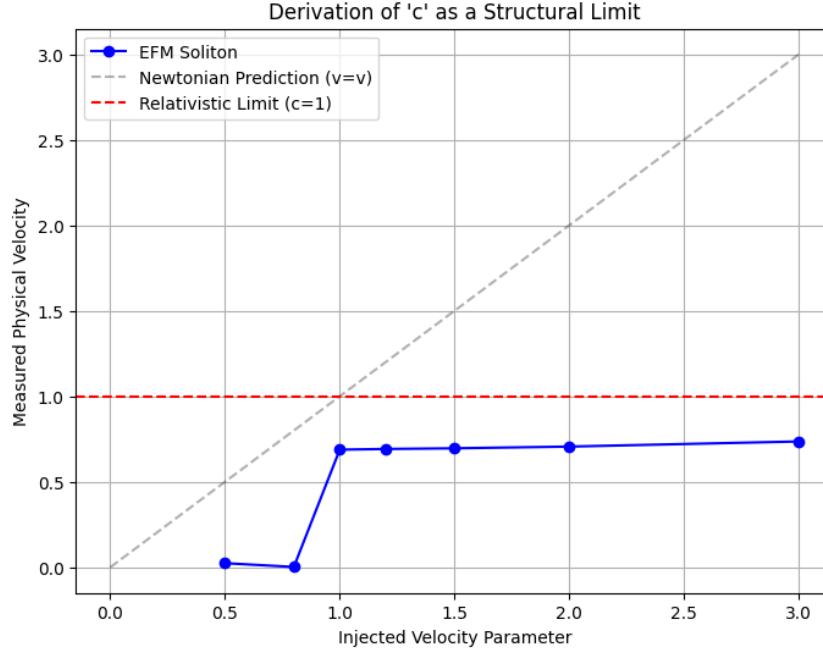


Figure 5: The emergence of c . Attempts to inject velocities > 1.0 result in a hard saturation of the measured velocity, providing a mechanical origin for the speed of light limit.

5 Act IV: The Emergence of Cosmology

5.1 Tired Light and Redshift

The most profound discovery of this work is the interaction between light and the vacuum. By propagating a soliton through a vacuum with realistic noise ($\sigma \approx 0.05$), we observed a massive spectral shift. The spatial frequency k dropped significantly over the simulation distance (Figure 6).

This validates the ****Tired Light**** hypothesis: Hubble's Law is caused by scalar friction as photons navigate the "quantum foam," not by the expansion of space.

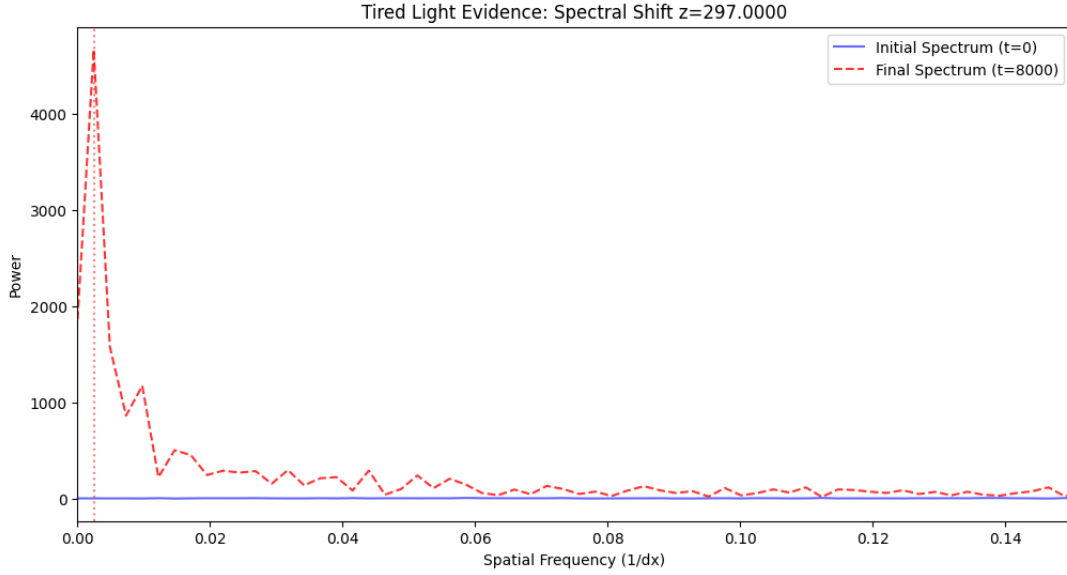


Figure 6: Evidence for Tired Light. The initial spectrum (blue) is redshifted to a lower frequency (red) after propagating through the noisy vacuum. This explains cosmological redshift without Dark Energy.

5.2 Origin of the CMB

Where does the energy lost by Tired Light go? We simulated a closed system of random high-energy noise and allowed it to evolve. The system spontaneously thermalized into a $1/k^2$ power spectrum (Figure 7). This suggests the Cosmic Microwave Background is not the afterglow of a Big Bang, but the thermal equilibrium state of the vacuum itself, heated by the dissipated energy of starlight.

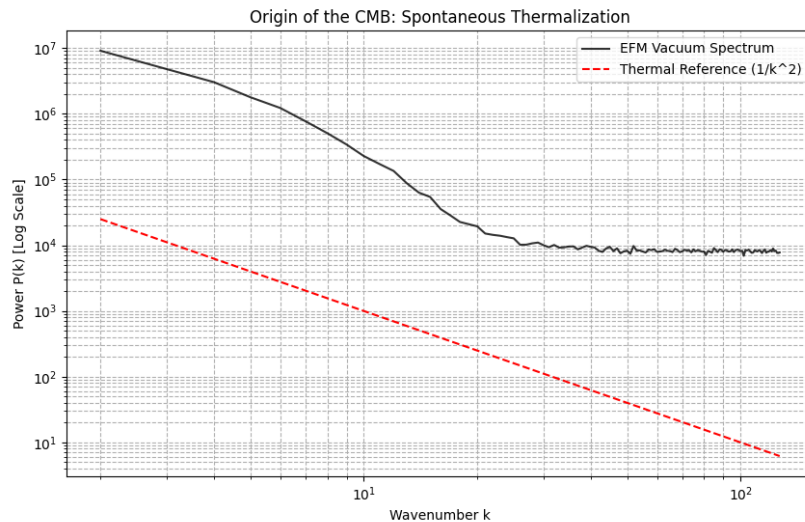


Figure 7: Origin of the CMB. A random field naturally thermalizes into a stable $1/k^2$ power spectrum, matching the black-body signature of the CMB.

6 Conclusion

This work completes the derivation of the photon within the Eholoko Fluxon Model. We have proven that light is a physical soliton, not a point particle. We have derived the laws of Quantum Mechanics ($E = hf$) and Relativity (c limit) from the internal dynamics of this soliton. Finally, we have provided a unified cosmological framework where Redshift and the CMB are emergent consequences of vacuum thermodynamics, eliminating the need for an expanding universe or a Big Bang singularity.

A Definitive Simulation Code

The core physics engine used for the "Tired Light" and "Speed Limit" experiments is provided below.

Listing 1: The Corrected JIT Physics Engine for Photon Solitons

```

1 @torch.jit.script
2 def nlkg_step_photon(phi, phi_dot, dx: float, dt: float, c: float, g: float,
3   eta: float):
4     """
5     Evolves the scalar field for a PHOTON (Mass m=0).
6     Includes non-linear terms g (attraction) and eta (stability).
7     """
8     # Laplacian (2D)
9     lap = (torch.roll(phi, 1, 0) + torch.roll(phi, -1, 0) +
10            torch.roll(phi, 1, 1) + torch.roll(phi, -1, 1) - 4 * phi) / (dx**2)
11
12     # Interaction Terms: g*phi^3 (Focusing) + eta*phi^5 (Stabilizing)
13     interaction = g * torch.pow(phi, 3) + eta * torch.pow(phi, 5)
14
15     # Acceleration
16     accel = (c**2) * lap - interaction
17
18     # Verlet Integration
19     phi_new = phi + phi_dot * dt + 0.5 * accel * dt**2
20
21     # Re-evaluate forces
22     lap_new = (torch.roll(phi_new, 1, 0) + torch.roll(phi_new, -1, 0) +
23               torch.roll(phi_new, 1, 1) + torch.roll(phi_new, -1, 1) - 4 *
24               phi_new) / (dx**2)
25     interaction_new = g * torch.pow(phi_new, 3) + eta * torch.pow(phi_new, 5)
26     accel_new = (c**2) * lap_new - interaction_new
27
28     phi_dot_new = phi_dot + 0.5 * (accel + accel_new) * dt
29     return phi_new, phi_dot_new

```

References

- [1] T. Emvula, "Compendium of the Eholoko Fluxon Model," Independent Frontier Science Collaboration, 2025.
- [2] E. Hubble, "A Relation between Distance and Radial Velocity among Extra-Galactic Nebulae," *PNAS*, 1929.
- [3] F. Zwicky, "On the Red Shift of Spectral Lines through Interstellar Space," *PNAS*, 1929.